CLOSED IMMERSIONS ARE PROPER

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Base change of a closed immersion is a closed immersion, and closed immersions are separated. Schemes, Lemma \ref{schemes-lemma-base-change-immersion} and Schemes, Lemma \ref{schemes-lemma-immersions-monomorphisms}, cited at proof lines 9938-9942. Give universal closedness and separatedness.
- A closed immersion is of finite type. Lemma \ref{lemma-immersion-locally-finite-type}, cited at proof lines 9943-9944. Gives the third ingredient.

Source reference. Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01JY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2031–2031. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01L7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2076–2076. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01T5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2362–2362. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *A closed immersion is proper, hence a fortiori universally closed.*

Proof. The base change of a closed immersion is a closed immersion (Schemes, Lemma 01JY). Hence it is universally closed. A closed immersion is separated (Schemes, Lemma 01L7). A closed immersion is of finite type (Lemma 01T5). Hence a closed immersion is proper. □